

How the Virtual Office Operates

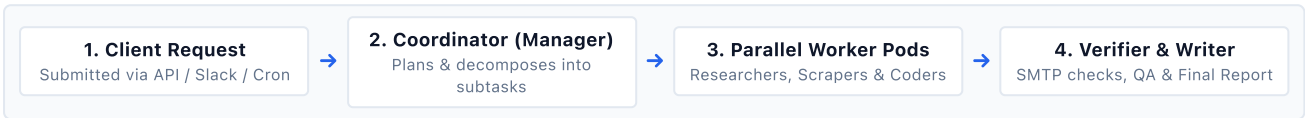
The Conceptual Architecture of an Autonomous Multi-Agent Organization

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The Virtual Office Hierarchy: Division of Labor in Action

Fathom does not operate as a single monolithic prompt. It structures work like an agile digital consulting agency where a Coordinator (Manager) delegates to specialized Worker Pods, supervised by Analysts and formatted by Writers.

Organizational Workflow Topology



1. COORDINATOR AGENT

Analyzes the objective, sets token budgets, establishes task trees, and tracks progress across child sub-agents with fair-share scheduling.

2. PARALLEL WORKERS

Execute discrete searches, scrape registries, run Python data transformations, and operate web browsers simultaneously across CPU threads.

3. QUALITY JUDGE & WRITER

Validates data completeness against the initial goal, triggers gap-filling rounds, and compiles clean deliverables into PDF/Excel.

Context Protection & Disk-Spill Backing Architecture

Executive Summaries Only: Sub-agents return concise executive findings (200–500 tokens) to the coordinator manager.

Raw Disk Spill: Gigabyte-scale raw HTML, JSON payloads, and scrape dumps are written to disk workspaces (~/.fathom/jobs/), keeping LLM context clean.

Enterprise Impact: High-level managers never experience context overflow or degraded reasoning quality, even during massive 1,000-page research runs.